

Generated Question Paper

Q1. (b) (i) Name the source of energy used by organism A. [1 marks]

Q2. (iii) Ancestors of monkey flower plants had a much smaller leaf area. Explain how the monkey flower plants have developed a larger leaf area over time. [5 marks]

Q3. The effect of pH on lipase activity is shown on Fig.1. [6 marks]

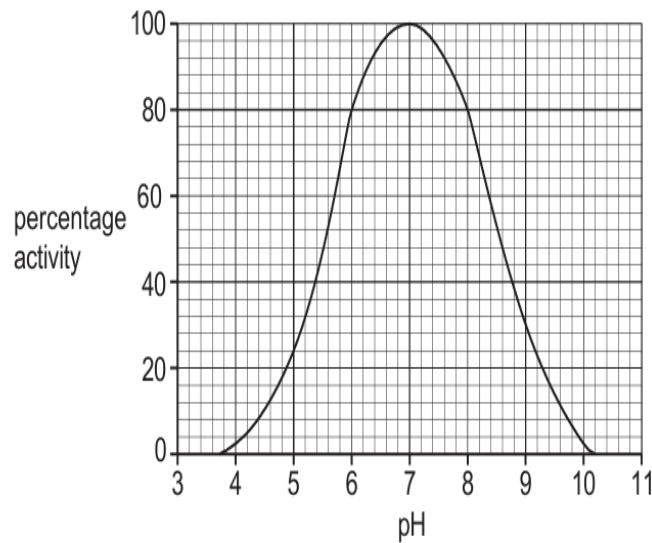


Figure: Fig1_1_LipaseActivity.png

Q4. The advantage of using a pyramid of energy is that it is more efficient. [2 marks]

Q5. (f) Discuss the effects of non-biodegradable plastics on terrestrial ecosystems. [5 marks]

Q6. The a) The possible causes of increased atmospheric carbon dioxide. [3 marks]

Q7. (ii) Enzymes are the chemical elements found in all proteins. [2 marks]

Q8. The testes are a part of the human body. Meiosis is the process of dividing the testes into two parts. [3 marks]

One has been completed for you.

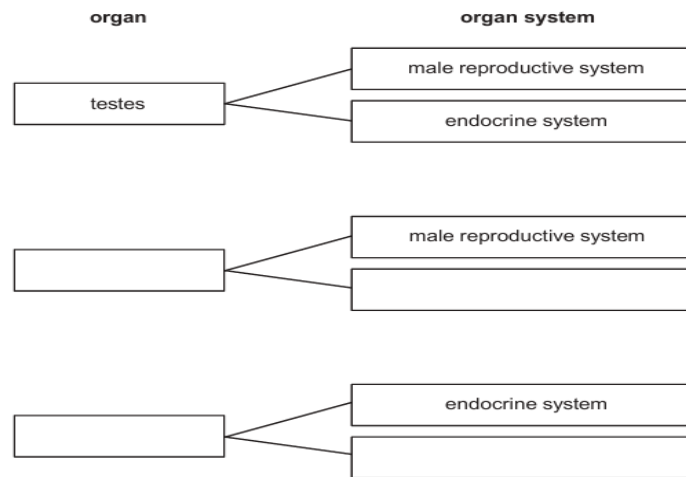


Fig. 5.1

Figure: organs_two_systems.png

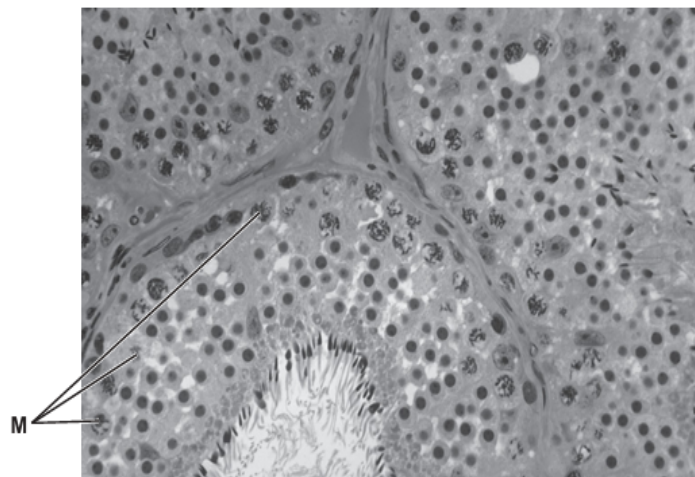


Fig. 5.2

Figure: mammalian_testis_micrograph.png

Q9. The following are the advantages of this method: [3 marks]

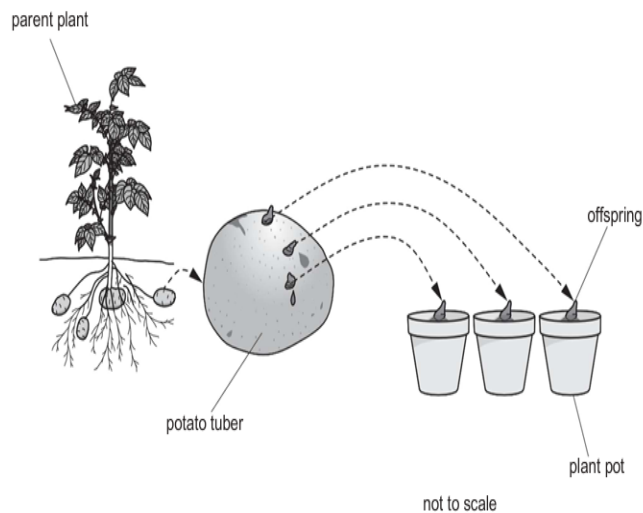


Fig. 6.2

Figure: potato_tubers_method.png

Q10. Plants store starch in their leaves. [2 marks]

Q11. (d) Explain why reflexes occur in unconscious people. [1 marks]

Q12. The process of converting ammonium ions into nitrate ions is called the nitrate reaction. [1 marks]

Q13. The average photosynthesis rate of soybean plants was measured in two plots with different CO₂ concentrations, 370 ppm and 550 ppm. Describe and explain the effect of carbon dioxide concentration on the average rates of photosynthesis of the soybean plants from 04:00 to 22:00. [6 marks]

The atmospheric carbon dioxide concentrations in the two plots were kept at:

- 370 ppm, which is similar to the current atmospheric carbon dioxide concentration
- 550 ppm, which is a possible future atmospheric carbon dioxide concentration.

When the soybean plants were fully grown, scientists calculated the average rates of photosynthesis at regular intervals from 04:00 to 22:00 for both plots.

The results are shown in Fig. 2.1.

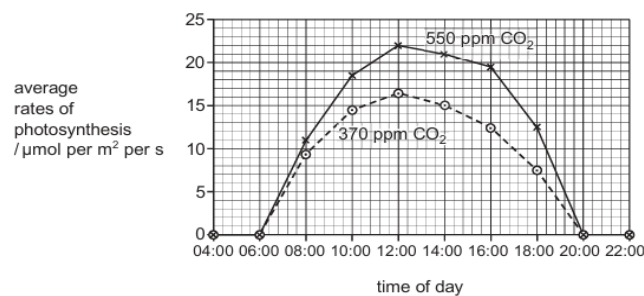


Fig. 2.1

Figure: photosynthesis_rate_graph.png